YONSEI UNIVERSITY

Tabular Data Analysis

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Severance

What's discussed in the last meetings

- 인체 임상 변수

Variable	Description	Units / Classification
File Analyzed	환자 / 케이스 식별자	ID
Osteoporosis	골다공증 여부 (0=정상, 1=골다공증)	Binary
Age	나이	세
진료년월일	검사 날짜	YYYY-MM-DD
키	키	cm
체중	몸무게	kg
BMI	체질량지수	kg/m²
허리둘레	허리 둘레	cm
근육량SoftLeanMass	부위별 평균 근육량	kg
체지방율BodyFat	체지방 비율	%
체지방량FatMassKg	체지방 총량	kg
복부지방율WHRatio	복부지방률 (WH Ratio)	%
복부내장지방량	내장지방량	kg
복부피하지방량	피하지방량	kg
초경연령ALL	초경 시작 나이	세
폐경여부	폐경 여부 (1=예, 0=아니오)	Binary
출산횟수ALL	출산 횟수	Count

- 영상 파생 변수

Variable	Description	Units / Classification
BreastArea(sqcm)AVG	유방 전체 면적 평균	cm²
DenseArea(sqcm)AVG	치밀 조직 면적 평균	cm²
BreastDensity(%)AVG	유방 내 치밀조직 비율	%
Breast_Density_Grading	유방 밀도 등급 (1~4단계)	Categorical
BIRADS_Category	BI-RADS 평가 카테고리	1~5 등급

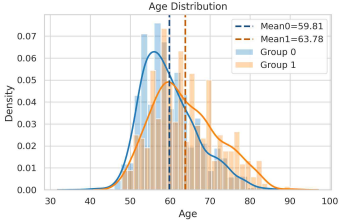
Group	Samples
Total	3,961(100)
Normal	3,475(87.7)
Osteoporosis	486(12.3)

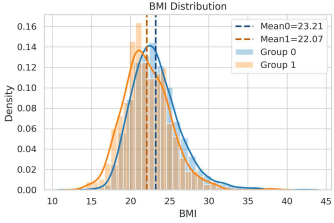
What's discussed in the last meetings

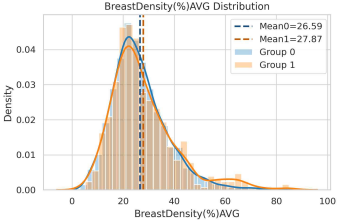
Variable	Classification	N_Sample	mean	std	min	25%	50%	75%	max
Age	Total	3,961	60.3	7.34	36	55	59	65	90
	Normal(0)	3,475	59.81	7.07	36	55	59	64	86
	Osteoporosis(1)	486	63.78	8.18	48	58	62	69	90
BMI	Total	3,961	23.07	3.12	12.5	20.9	22.7	24.8	42.2
	Normal(0)	3,475	23.21	3.12	12.5	21.1	22.9	24.9	42.2
	Osteoporosis(1)	486	22.07	2.99	14.3	20.1	21.8	24.0	38.1
BreastDensity(%)AVG	Total	3,961	26.75	11.45	3.13	19.26	24.81	32.28	89.30
	Normal(0)	3,475	26.59	11.24	3.13	19.18	24.78	32.15	89.30
	Osteoporosis(1)	486	27.87	12.81	5.12	19.78	24.96	33.53	84.95
BreastArea(sqcm)AVG	Total	3,961	122.54	38.10	27.45	95.33	117.25	144.23	377.72
	Normal(0)	3,475	123.57	38.38	27.45	96.75	118.56	145.10	377.72
	Osteoporosis(1)	486	115.22	35.22	31.55	88.61	110.13	137.10	241.37
DenseArea(sqcm)AVG	Total	3,961	29.73	10.59	9.35	22.49	27.81	34.68	97.78
	Normal(0)	3,475	29.87	10.71	9.35	22.56	27.90	34.80	97.78
	Osteoporosis(1)	486	28.68	9.58	10.04	21.72	27.01	33.87	66.51

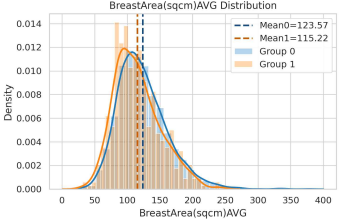
Type	Age	BMI	BreastDensity(%)AVG	BreastArea(sqcm)AVG	DenseArea(sqcm)AVG
t-stat	-10.18	7.86	-2.09	4.84	2.53
p-value	<0.001	<0.001	0.037	<0.001	0.012

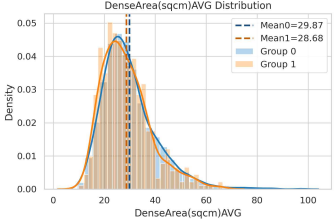
What's discussed in the last meetings

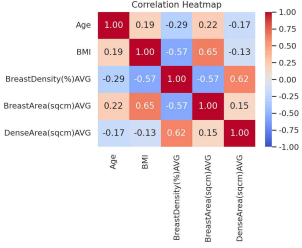












Research Summary

- **Postmenopausal Cohort Analysis (Logistic Regression)**
- **Evaluated metrics:** Accuracy, Precision, Recall, F1 score, ROC AUC
- Compared Data oversampling with y=1 with **SMOTE method**
- Compared **Holdout vs. k-fold Cross-Validation** → similar performance across all metrics
  - Conducted hyperparameter tuning via Grid Search (param_grid)
    - Higher performance with lower precision code improvements leads to lower Recall
  - Performed **stratified analyses using Age, BMI, BDG**
    - **Age:** the older the age, the higher the recall
    - **BMI:** the lower the number, the higher the recall
    - **BDG:** in Grade 2, recall is highest

Detailed Progress

- **Dataset Split**
  - **폐경군 Total 3961개에 대해서 8:2로 Train:Test로 Dataset Split 진행**
  - **Train Dataset : 3168(2779/389), Test Dataset : 793(696/97), scale_pos_weight=7.144**
    - stratify 옵션을 사용하여 Dataset 비율 유지
- **Pipeline**

```python
def build_model():
    return Pipeline([
        ("scaler", StandardScaler()),
        ("clf", LogisticRegression(
            max_iter=1000,
            class_weight="balanced",
            solver="liblinear"
        ))
    ])
```

Detailed Progress

- **Logistic Regression Results**
  - **Test Metrics : 793 [696 / 97]**

Feature Set	Accuracy	Precision	Recall	F1	AUC
[age, bmi]	.6595	.2164	.6804	.3284	.7144
[age, bmi, bda]	.6507	.2097	.6701	.3194	.7105
[age, bmi, bdg]	.6608	.2171	.6804	.3292	.7146
[age, bmi, da]	.6545	.2117	.6701	.3218	.7051

  - **SMOTE | Test Metrics : 793 [696 / 97]**

Feature Set	Accuracy	Precision	Recall	F1	AUC
[age, bmi]	.6608	.2171	.6804	.3292	.7148
[age, bmi, bda]	.6545	.2136	.6804	.3251	.7106
[age, bmi, bdg]	.6633	.2167	.6701	.3275	.7114
[age, bmi, da]	.6507	.2078	.6598	.316	.7066

Detailed Progress

- **Grid Search**

```
param_grid = {
    "clf__C": [0.01, 0.1, 1, 10],
    "clf__penalty": ["l1", "l2"],
    "clf__solver": ["liblinear"],
    "clf__class_weight": [None, "balanced"],
}
```

- **Compare Versions**
  - **Holdout**
  - **Holdout (GridSearch)**
  - **5-fold CV (fixed params)**
  - **5-fold CV (GridSearch)**

Detailed Progress

- **Logistic Regression Results**
  - **Test Metrics : 793 [696 / 97]**

Feature Set	Method	Accuracy	Precision	Recall	F1	AUC
[age, bmi]	Holdout	.6595	.2164	.6804	.3284	.7144
	5-fold CV (fixed params)	.6595	.2164	.6804	.3284	.7144
	Holdout (gridSearch)	.8789	1.0	.0103	.0204	.7148
	5-fold CV (gridSearch)	.8777				.7117
[age, bmi, bda]	Holdout	.6507	.2097	.6701	.3194	.7105
	5-fold CV (fixed params)	.6507	.2097	.6701	.3194	.7105
	Holdout (gridSearch)	.8789	1.0	.0103	.0204	.7118
	5-fold CV (gridSearch)	.8777				.7148
[age, bmi, bdg]	Holdout	.6608	.2171	.6804	.3292	.7146
	5-fold CV (fixed params)	.6608	.2171	.6804	.3292	.7146
	Holdout (gridSearch)	.8789	1.0	.0103	.0204	.7118
	5-fold CV (gridSearch)	.8777				.7148
[age, bmi, da]	Holdout	.6545	.2117	.6701	.3218	.7051
	5-fold CV (fixed params)	.6545	.2117	.6701	.3218	.7051
	Holdout (gridSearch)	.8789	1.0	.0103	.0204	.7118
	5-fold CV (gridSearch)	.8777				.7056

Detailed Progress

- **Stratified Analysis**
  - **Age : <60, 60~69, >=70**
  - **BMI : <18.5, 18.5~22.9, 23~24.9, >=25**
  - **Breast_Density_Grading : 1, 2, 3, 4**

Detailed Progress

- **Logistic Regression Results**
  - **Test Metrics : 793 [696 / 97]**

Feature Set	Stratified Variable	Method	N	Accuracy	Precision	Recall	F1	AUC
[age, bmi]	Age	<60	417	.8201	.1719	.3333	.2268	.6628
		60-69	276	.5543	.2282	.8095	.3560	.7126
		>=70	100	.2800	.2283	**.9545**	.3684	.6696
	BMI	<18.5	36	.4444	.3667	**.9167**	.5238	.5729
		18.5-22.9	378	.5582	.2083	.7273	.3239	.6748
		23-24.9	197	.7766	.1957	.5625	.2903	.6708
		>=25	182	.7857	.1622	.4286	.2353	.7085
[age, bmi, bda]	Age	<60	417	.8153	.1562	.3030	.2062	.6647
		60-69	276	.5362	.2208	.8095	.3469	.7070
		>=70	100	.2800	.2283	**.9545**	.3684	.6696
	BMI	<18.5	36	.4444	.3571	**.8333**	.0500	.5764
		18.5-22.9	378	.5423	.2020	.7273	.3162	.6730
		23-24.9	197	.7716	.1915	.5625	.2857	.6744
		>=25	182	.7857	.1622	.4286	.2353	.7015

Detailed Progress

- **Logistic Regression Results**
  - **Test Metrics : 793 [696 / 97]**

Feature Set	Stratified Variable	Method	N	Accuracy	Precision	Recall	F1	AUC
[age, bmi, bdg]	Age	<60	417	.8225	.1746	.3333	.2292	.6634
		60-69	276	.5543	.2282	.8095	.3560	.7129
		>=70	100	.2800	.2283	**.9545**	.3684	.6684
	BMI	<18.5	36	.4444	.3667	**.9167**	.5238	.5729
		18.5-22.9	378	.5608	.2094	.7273	.3252	.6755
		23-24.9	197	.7766	.1957	.5625	.2903	.6721
		>=25	182	.7857	.1622	.4286	.2353	.7077
	BDG	1	54	.7222	.2143	.4286	.2857	.6413
		2	89	.6180	.2979	**.9333**	.4516	.8428
		3	540	.6537	.1850	.6066	.2835	.6752
		4	110	.7000	.2791	.8571	.4211	.7984
[age, bmi, da]	Age	<60	417	.8106	.1714	.3636	.2330	.6669
		60-69	276	.5507	.2230	.7857	.3474	.7010
		>=70	100	.2900	.2247	**.9091**	.3604	.6358
	BMI	<18.5	36	.3889	.3438	**.9167**	.5000	.5625
		18.5-22.9	378	.5450	.2030	.7273	.3175	.6705
		23-24.9	197	.7817	.1860	.5000	.2712	.6671
		>=25	182	.7967	.1714	.4286	.2449	.7020

Next Plan

- **Check results and improve performance with other models (XGBoost)**
- **Deep learning performance evaluation by combining tabular data and DICOM**

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